



**NAZARBAYEV
UNIVERSITY**

BUS 101 - Core Course in Business

Final Project Report

AgroKomek

May 2, 2025

Prepared by: Aknur Zhakanov, Alen Kazhymukan, Aiym Madil, Madiyar Saduakas

Alima Nurgaliyeva, Yersain Sarbassov, Sanzhar Ussin

The problem

The absence of an online place to communicate with/between farmers is the problem that “Agrokomek” will address by creating a user-friendly online app. Agriculture is an important sphere in Kazakhstan's economic development. It is hard for farmers to find other farmers in order to exchange the cattle. Moreover, people who are willing to buy cattle also suffer from the absence of variability of where they can buy.

Aim of the project

We are developing a mobile app designed to serve as a networking platform for farm owners. The app will feature an interactive map showing the live locations of other users, similar to the friend-tracking feature in 2GIS. Users will be able to chat with each other, share suggestions with the government, and stay informed through a dedicated news feed focused on agriculture and political developments that may impact farmers.

Additionally, the app will provide a convenient way for individuals interested in purchasing livestock to locate nearby farms, contact owners directly, and negotiate deals online

Value proposition

1. Find neighbours

As was said before, the opportunity to find the position of the neighbouring farms for us is the main aim of the project.

2. Create a market for buying the cattle

All farmers will be able to put their cattle up for sale at the price they wish. People who want to buy cattle will have options and the ability to negotiate with owners, the ability to find the closest farms and so on. People in Kazakhstan may buy cattle for

multiple reasons, such as religious reasons (Kurban ait, sacrifice, etc), for personal use (like horses)

Stronger Relationship

We will create a community which is united by one common topic: agriculture.

People will be able to share their knowledge with each other, providing valuable insights, increasing the overall level of knowledge in this sphere.

3. Scalable tech

We will have a team of developers who will constantly work on updates of the app, based on the feedback of users.

Products and services

The main product that we offer is the innovative mobile app named “Agrokomek”. It will be free to download from all app markets like “Apple App Store”, “Google Play Market” and so on. In this app, users will be provided with numerous services listed below:

- Always-on online map with marked locations of other users (farmers), similar to 2GIS friends. Users, equivalently, farmers will register and verify the existence of the farm, put a fixed marker for their location of the farm, and it will be shown to every user in the app.
- Built-in online catalogue with the ability to talk to owners of cattle, similar to kolesa.kz or kaspi.kz online shop. Users who want to buy from farmers, whether it other owners of a farm or just a person willing to buy cattle, will be able to see tons of offers.
- Users will be provided with everyday news inside the app. All governmental changes, forums and information related to agriculture will be posted regularly.

If a user registers as a farm owner, they should pay a yearly subscription that costs 5000 tenge (approximately 10 dollars), otherwise, the location of their farm will not be visible to others. Users who want to access the market of cattlemarket can do it on a free basis.

The strategy of Getting Our Products to Customers

We use a digital-first approach to successfully reach our target customers who consist of Kazakhstani farmers. Agrokomek will become available on the Apple App Store and Google Play Market platforms during the first annual period. Our product distribution strategy includes digital marketing directed at farmers supported by cooperative organization and agricultural union collaborations.

The app promotion strategy includes online platforms with rural farming events along with national rural development programs. Our primary attention right now targets users from Akmola together with Turkestan and East Kazakhstan regions to establish a critical user mass before we proceed with expanding our service toward every rural area. Local akimats will join forces to host offline workshops and demo sessions to assist farmers on how to use our app.

Gradual Expansion with Growing Market Demand

The expansion strategy of Agrokomek corresponds to Kazakhstan's digital agriculture direction, while rural residents adopt smartphones in increasing numbers. The expansion will begin by utilizing farmer associations with existing government infrastructure as well as agricultural exhibitions, to gradually develop a dependable user network. We expect customer demand for enhanced functionality to accelerate once trust spreads naturally and broadens its consumer base. Especially from vendors who seek distribution platforms, from agri-tech startups who want to implement their APIs and data tools, and from meat buyers who are interested in direct access to local farms.

The estimated market size

Target Market: 240,720 registered farm entities in Kazakhstan (The Bureau of National Statistics, 2023).

We assume that at least 18% of farm owners, which is ~43,000 farms, have the potential to adopt Agrokomek because they own medium—to large-scale farms and use digital technology extensively. Agrokomek aims to acquire 9,845 active users in Year 1 followed by a continuous expansion of this user base:

Estimated Subscription Sales: 9,845 (total sold, per year).

Average Selling Price: 5,000 tenge (average subscription cost).

Market Size Calculation: Estimated Subscriptions $\times$ Average Selling Price = $9,845 \times 5,000 = 49,225,000$ tenge per year.

Market Share

It is crucial to analyze and calculate the market share and further plan a market strategy to successfully launch the business. Based on the data above, our goal is to take a stable market share in the Kazakhstani market within five years of operation.

1. Market Share Projections.

Market share estimation includes considering our sales objectives and projected industry revenue levels.

We have made annual projections based on steady company advancement and an increasing client base as follows:

- Year 1: Establish presence, targeting a ~4% market share.
- Year 2: Expanding marketing and partnerships, aiming for a ~8% market share.
- Year 3: Strengthening reputation and customer base, achieving a ~12% market share.
- Year 4: Expanding the product line and launching premium goods to increase market share to ~15%.
- Year 5: Maximizing market penetration through strategic innovations, achieving a ~19% market share.

2. Sales Overview

Pricing and Sales Figures

AgroKomek follows a freemium model designed to attract a wide audience of Kazakhstani farmers. The projected growth in Premium users over five years is shown below:

Package	Price	Year 1	Year 2	Year 3	Year 4	Year 5
---------	-------	--------	--------	--------	--------	--------

Premium	₹5,000	9485	18881	28189	37411	46550
---------	--------	------	-------	-------	-------	-------

Table 1. Total Premium Subscribers

- **Year 1:** 9,485 users
- **Year 5:** 46,550 users

3. Revenue Breakdown

Revenue is driven by Premium subscriptions. Assuming an average of ₹5,000 per subscriber per month:

Year	Revenue
Year 1	₹47,425,000
Year 2	₹94,405,000
Year 3	₹140,945,000
Year 4	₹187,055,000
Year 5	₹232,750,000

Table 2. Premium Revenue Projections

4. Cost Structure

Our cost structure includes both the cost of goods sold (COGS) related to app services and operating expenses.

Cost of Goods Sold (COGS)

COGS includes expenses for servers and app maintenance, which are tied to the number of users.

Category	Year 1	Year 2	Year 3	Year 4	Year 5
Servers	₹2,845,500	₹6,060,801	₹8,456,700	₹11,223,300	₹13,965,000
Maintain of app	₹3,699,150	₹7,879,041	₹11,763,270	₹15,611,610	₹19,425,315
Total	₹6,544,651	₹13,939,844	₹20,219,973	₹26,834,914	₹33,390,320

Table 3. Cost of Goods Sold

Fixed and Variable Costs (Operating Expenses)

Operating expenses include salaries, depreciation, rent, travel, amortization of app creation, and advertising, are projected as follows:

Category	Year 1	Year 2	Year 3	Year 4	Year 5
Salaries	₹15,408,000	₹16,486,560	₹22,050,774	₹28,313,194	₹35,344,304
Depreciation	₹24,000	₹24,000	₹696,000	₹1,040,000	₹1,384,000
Rent	₹7,200,000	₹7,905,600	₹8,680,349	₹9,531,023	₹10,465,063
Travel	₹16,560,000	₹17,719,200	₹18,959,544	₹19,528,330	₹20,114,180
Amortization of app creation	₹24,000,000	₹24,000,000	₹24,000,000	₹24,000,000	₹24,000,000
Advertisement	₹5,000,000	₹5,350,000	₹5,724,500	₹6,125,215	₹6,553,980
Total of operating expenses	₹68,192,000	₹71,485,360	₹80,111,167	₹88,537,762	₹97,861,527

Table 4. Operating Expenses

Profit Margins

The net income shows the profitability of the project over the five years.

Year	Gross Profit	Operating Expenses	Net Income
1	₹40,880,349	₹68,192,000	-₹24,580,486
2	₹80,465,156	₹71,485,360	₹8,081,816
3	₹120,725,027	₹80,111,167	₹36,552,474
4	₹160,220,086	₹88,537,762	₹64,514,091
5	₹199,359,680	₹97,861,527	₹91,348,338

Table 5. Profit Margins

Workforce and Resource Allocation

The team structure ensures operational efficiency:

Position	Year 1	Year 5
Software Developer	1	1
IT support	1	4
External affair Manager	1	1
Accountant	1	1
Content Creator	1	1
Customer Support	1	2

Table 6. Workforce Structure

To support user growth, our team will expand over five years. This includes support reps for onboarding and developers for backend scalability. Salaries reflect increasing headcount and inflation. App creation costs are amortized evenly each year.

Priorities for Action:

In order to meet our growth goals and generate early momentum, we have the following list of priorities.

1. **Complete MVP development and UX testing** by Q2.

The minimum viable product (MVP) will include core features like live map, chat functionality, cattle listing and profile page. Small focus group of farmers will be invited to test the app on its user friendliness under real conditions and collect feedback on our app.

2. **Collaborate with regional agricultural departments** for implementing the app implementation and its promotional activities.

The partnerships will enable us to establish trust with communities as we acquire their geographical access. Local authorities provide help to schedule educational campaigns together with promotional content distribution and could involve AgroKomek in region-wide digital agricultural projects.

3. **Launch pilot campaigns** in 3 regions by Q3.

We will select target regions with high agricultural activity and a mix of traditional and modern farms such as Akmola, East Kazakhstan, and Turkestan regions. Our marketing approach combines digital ads with public protests alongside famous farmer representatives who will promote the app.

4. **Design referral incentives** to encourage viral user acquisition.

The implementation of incentive plans will stimulate users to refer others through viral acquisition. People who refer new users to join the application will gain benefits including premium account access for free along with additional listing visibility features. The incentive program aims at building our user base through viral growth initiatives at affordable marketing expenses.

5. **Recruit support reps and devs** for onboarding, localization, and technical scalability.

The team requires new hires for support representatives and developers to provide onboarding as well as localization and technical scalability. New users who need assistance obtaining basic understanding of mobile technology will be supported by representatives while developers will work on expanding the backend infrastructure and extending app functionality across different connectivity settings.

These actions will develop a unified system where local farmers establish relationships and deal with each other and collaborate thus transforming the entire agricultural sector of Kazakhstan.

Key assumptions

AgroKomek represents an essential opportunity to solve the problem of the connectivity gap in Kazakhstan's agricultural sector. We assume that there is a strong demand for a centralized platform that provides farmer networking, livestock trading, and access to agricultural news, because of the inefficiencies of current methods like word-of-mouth or outdated catalogs. It is expected that the interactive map, accessible chat functionality and marketplace will appeal to the farmers who are looking for a comfortable solution. The user-friendly design of the app is created for various levels of digital literacy, making it accessible.

The business model is assumed to be scalable, with low incremental costs to have an opportunity to add new users. Revenue streams, such as transaction fees from livestock sales or premium subscriptions for advanced features (e.g., analytics or priority listings), are expected to provide high margins. Partnerships with agricultural organizations and government will enrich credibility and trust of the users, which are the most important to adopt the app. The opportunity to share opinions, negotiate deals, and access news will be the main advantage to repeat the use of app. The long-term goal is to expand further into other agricultural services, such as tools for crop management and equipment trading.

Finally, we consider that the competitive landscape for agricultural networking apps in Kazakhstan is not yet developed sufficiently, giving our platform an advantage of a first-mover.

Summary Financial Projections and Expected Returns to Investors

Prices

Our business model is based on a freemium approach, with optional premium subscription. The pricing structure is based on market research on similar platforms and the purchasing power of farmers in Kazakhstan. Two subscription tiers are proposed:

- **Basic (Free):** Includes access to the interactive map, basic chat, and catalog. This subscription is intended for cost-conscious farmers.
- **Premium (5,000 KZT/month):** Offers advanced features like priority listings, analytics on market trends, and access to a dedicated news feed with agricultural insights. This subscription is intended for large companies or regular traders.

Cost of Goods Sold (COGS)

The Cost of Goods Sold includes expenses of the app's services, such as server costs and app maintenance. Estimated at about 14 % annually. These costs increase depending on the number of subscribers and are projected as follows:

Years	1	2	3	4	5
Servers	₹2,845,500	₹6,060,801	₹8,456,700	₹11,223,300	₹13,965,000
Maintain of app	₹3,699,150	₹7,879,041	₹11,763,270	₹15,611,610	₹19,425,315
Total	₹6,544,651	₹13,939,844	₹20,219,973	₹26,834,914	₹33,390,320

Table 7. Cost of Goods Sold.

Expenses

Operating expenses are an essential component of the business's financial structure. These include salaries, which start at (15,408,000 KZT) in Year 1, rising annually due to team expansion and inflation adjustments, reaching (35,344,304 KZT) by Year 5. Rent expenses begin at (7,200,000 KZT) in Year 1 and increase by 10% yearly, rising to (10,465,063 KZT) in Year 5. Travel expenses, important for partnership development and customer outreach, grew from (16,560,000 KZT) in Year 1 to (20,114,180 KZT) in Year 5, reflecting increased operational activity. The amortization of app creation costs, a fixed annual expense, remains steady at (24,000,000 KZT) for all five years. Advertising costs start at (5,000,000 KZT) in Year 1, increasing with inflation and marketing efforts to (6,553,980 KZT) by Year 5. In sum, total operating expenses rose from (68,192,000 KZT) in Year 1 to (97,861,527 KZT) in Year 5. Capital expenditures (CapEx) include an initial investment of (120,000,000 KZT) in Year 0 for app development and fixed assets. The following years have smaller investments: (120,000 KZT) in Year 1, (3,360,000 KZT) in Year 3 for IT equipment upgrades, and (1,720,000 KZT) annually in Years 4–5 for maintenance. All these expenses are necessary to maintain operations and ensure the functioning of the business.

Initial Capital Requirements

Initial investment required to initiate the business includes capital spending (CapEx) on fixed assets such as a printer purchased in Year 1 for 120,000 KZT. There is no other fixed asset

purchase itemized for Year 1. Since the business has no other capital spending in the first year, overall initial investment required to initiate operations totals 120,000 KZT.

Products Number Growth

Agrokomek demonstrates strong and consistent growth in subscriber base, reflecting growing utilization of digital technologies in Kazakhstan's agriculture sector. Starting with 9,485 subscribers in Year 1, our foundation increased nearly two times by Year 2 to 18,881 subscribers. The growth continues steadily to 46,550. This growth trend reflects market demand and trust created through strategic partnerships, continued marketing, and enhanced app functionality as shown in the Table 8. By Year 5, Agrokomek is a vital platform for Kazakhstan farmers as they bring together nearly 47,000 users nationwide.

Subscription					
Years	1	2	3	4	5
Number of subscribers	9485	18881	28189	37411	46550
Revenue	₸47 425 000,00	₸94 405 000,00	₸140 945 000,00	₸187 055 000,00	₸232 750 000,00

Table 8. Predicted total number of customers provided in Excel Worksheet

Net Income

Agrokomek's finances show a clear transition from initial losses to strong profitability as the business scales. The Year 1 net loss of ₸24,580,486 reflects high up-front spending, namely the ₸24,000,000 amortization of app development and costly operating expenses such as salaries, marketing, and travel (Table 9). The initial loss was intentional, as in the beginning, focus was put on developing the app, obtaining a user base, and marketing in multiple regions.

From Year 2, the business is profitable at a net income of ₹8,081,816. This is driven by a sharp increase in subscribers (9,485 to 18,881) and subsequent revenue growth to ₹94,405,000. The greatest drivers of improved profitability are economies of scale and the relatively fixed nature of some costs (such as app amortization), which do not grow with the number of users.

By Year 5, net income is ₹91,348,338. Key drivers of this performance include consistent revenue growth from a growing subscriber base (46,550 users in Year 5) and the firm's ability to control growth in operational costs. Salaries and travel costs, in particular, grow modestly compared to revenue, enabling margin expansion.

Overall, the shift from loss to profit is largely attributable to a well-executed growth strategy that balances user acquisition with careful cost control. The upfront investment in marketing and technology sets the stage for long-term profitability with positive trends as shown in the *Figure 1. .*

Income Statement Operations	Year 1	Year 2	Year 3	Year 4	Year 5
Sales	₹47 425 000	₹94 405 000	₹140 945 000	₹187 055 000	₹232 750 000
COGS	-₹6 544 651	-₹13 939 844	-₹20 219 973	-₹26 834 914	-₹33 390 320
Gross profit	₹40 880 349	₹80 465 156	₹120 725 027	₹160 220 086	₹199 359 680
Operating expenses					
Salaries	-₹15 408 000	-₹16 486 560	-₹22 050 774	-₹28 313 194	-₹35 344 304
Depreciation	-₹24 000	-₹24 000	-₹696 000	-₹1 040 000	-₹1 384 000
Rent	-₹7 200 000	-₹7 905 600	-₹8 680 349	-₹9 531 023	-₹10 465 063
Travel	-₹16 560 000	-₹17 719 200	-₹18 959 544	-₹19 528 330	-₹20 114 180
Amortization of app creation	-₹24 000 000	-₹24 000 000	-₹24 000 000	-₹24 000 000	-₹24 000 000
Advertisement	-₹5 000 000	-₹5 350 000	-₹5 724 500	-₹6 125 215	-₹6 553 980
Total of operating expenses	-₹68 192 000	-₹71 485 360	-₹80 111 167	-₹88 537 762	-₹97 861 527
EBIT	-₹27 311 651	₹8 979 796	₹40 613 861	₹71 682 324	₹101 498 153
Taxes	₹2 731 165	-₹897 980	-₹4 061 386	-₹7 168 232	-₹10 149 815
Net Income	-₹24 580 486	₹8 081 816	₹36 552 474	₹64 514 091	₹91 348 338

Table 9. Income statement in Excel Worksheet

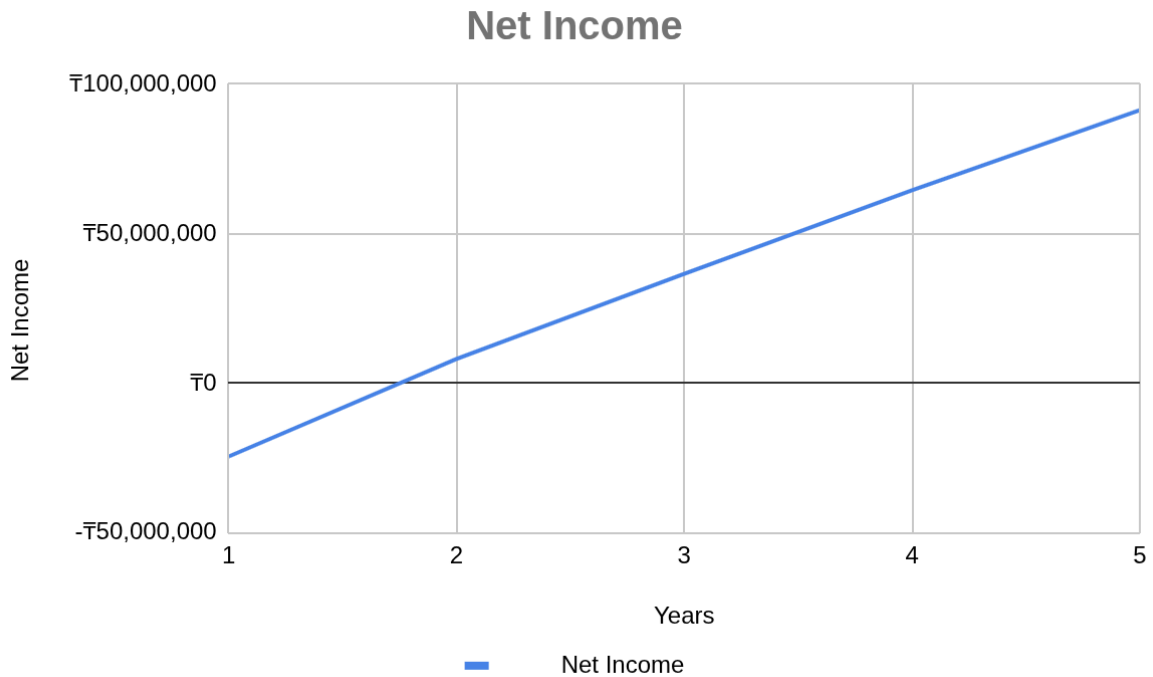


Figure 1. Net Income after taxes

Free Cash Flow

Agrokomek began with ₹120,000,000 initial investment in Year 0 that was used for the development of apps and capital investments that formed the basis of operations.

In Year 1, the Free Cash Flow (FCF) is -₹8,738,736, because of unavoidable expenses such as wages, advertising, and infrastructure. However, as subscription grows and revenue widens out, the company demonstrates a strong financial turnaround. In Year 2, Agrokomek registers positive FCF of ₹24,119,216, pointing towards long-term profitability. The trend continues to rise as FCF progresses to ₹49,976,674 in Year 3, ₹79,995,391 in Year 4, and even to ₹107,244,188 in Year 5 as shown in Table 10.

For long-term financial analysis, Free Cash Flow with Continuation Value (CV) was calculated using a 3% terminal growth rate. This results in a value in Year 5 of ₹757,017,795,

showing Agrokomek's ability in the future to continue profitability. With a Net Present Value (NPV) of ₸261,195,360 and an Internal Rate of Return (IRR) of 54%.

The positive and expanding cash flow trend reflects financial status, affirming optimism in the prosperity of Agrokomek and its position as a leader in Kazakhstan's emerging agri-tech industry of the future.

Operations	0	1	2	3	4	5
Unlevered Net Income		-₸24 580 486	₸8 081 816	₸36 552 474	₸64 514 091	₸91 348 338
Depreciation		₸24 024 000	₸24 024 000	₸24 696 000	₸25 040 000	₸25 384 000
Capex	-120000000	-₸120 000	-	-₸3 360 000	-₸1 720 000	-₸1 720 000
NWC		-₸8 062 250	-₸7 986 600	-₸7 911 800	-₸7 838 700	-₸7 768 150
FCF	-120000000	-₸8 738 736	₸24 119 216	₸49 976 674	₸79 995 391	₸107 244 188
Terminal Value						₸649 773 607
FCF with Terminal Value	-120000000	-₸8 738 736	₸24 119 216	₸49 976 674	₸79 995 391	₸757 017 795
DCF	-120000000	-₸7 282 280	₸16 749 456	₸28 921 687	₸38 578 024	₸304 228 473
NPV	\$261 195 359,96					
IRR	54%					
Payback Period	3,683074918					

Table 10. Operations in Excel Worksheet

Payback Period

Agrokomek's initial investment, for application development and required capital expenditures, will be recouped in entirety in approximately 3.7 years. This payback time is a sign of high subscriber growth and effective cost management, though the high initial investment in Year 0 extends the payback time somewhat beyond the 3-year mark. By mid-Year 4, overall net cash inflows will have exceeded overall startup and operating expenses, rendering the company able to operate profitably and sustainably from then on (Fig. 2).

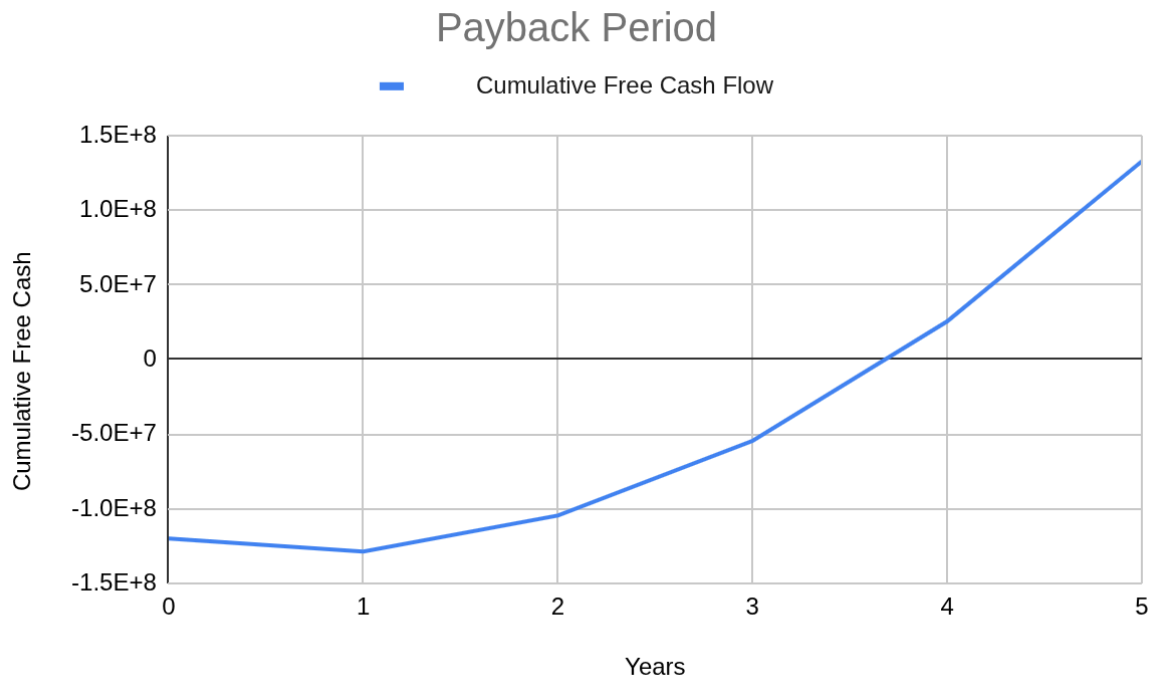


Figure 2. Payback Period

References

Bureau of National Statistics. (2023, February 6). *За год количество фермерских хозяйств в Казахстане выросло на 9%* [The number of farms in Kazakhstan increased by 9% over the year]. Eldala.kz.
<https://eldala.kz/novosti/kazahstan/13158-za-god-kolichestvo-fermerskih-hozyaystv-v-kazahstane-vyroslo-na-9>

We confirm that all members of our team have made a legitimate effort to contribute to this submission and agree that all members should receive the same Assessment Grade.